Circular Measure

Section 1: Radians, Arcs & Angles

1 Convert an angle of 150° into exact radian measure.

a. $\frac{5\pi}{6}$

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{7\pi}{6}$

2 A circle has a radius of 8 cm. Find the length of the arc intercepted by a central angle of 1.25 radians.

a. 10 cm

b. 6.4 cm

c. 12.5 cm

d. 8 cm

3 An arc of length 14π cm subtends a central angle of $\frac{7\pi}{4}$ radians. Find the radius of the circle.

a. 8 cm

b. 7 cm

c. 12 cm

d. 10 cm

Section 2: Sector Area & Perimeter Formulas

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4 Find the area of a sector of a circle with radius 6 cm and central angle $\frac{\pi}{3}$ radians.

a. 6π cm^2

b. 12π cm^2

c. 3π cm^2

d. 18π cm^2

5 A sector has a perimeter of 28 cm and radius 8 cm. What is the area of the sector in cm^2?

a. 48 cm^2

b. 56 cm^2

c. 64 cm^2

d. 32 cm^2

6 A wire of fixed length 36 cm is formed into a sector of a circle of radius r. What radius r maximizes the sector's area?

a. 9 cm

b. 6 cm

c. 12 cm

d. 8 cm

Section 3: Segments of Circles

7 A circle of radius 10 cm has a central angle of $\frac{\pi}{2}$ radians. Find the exact area of the minor segment.

a. $25\pi - 50$

b. $50\pi - 25$

c. $25\pi - 100$

d. $50\pi - 50$

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8 A chord of a circle of radius 12 cm subtends a central angle of $\frac{\pi}{3}$ radians. Find the exact length of the chord.

a. 12 cm

b. $6\sqrt{3}$ cm

c. $12\sqrt{3}$ cm

d. 6 cm

9 Find the perimeter of the minor segment cut off by a chord with central angle $\theta = \frac{2\pi}{3}$ in a circle of radius 6 cm.

a. $4\pi + 6\sqrt{3}$

b. $4\pi + 12$

c. $6\pi + 6\sqrt{3}$

d. $2\pi + 6\sqrt{3}$

Section 4: Compound Shapes & Exam Applications

10 Two identical circles of radius $r = 6$ cm pass through each other's centers. What is the area of the overlapping region?

a. $24\pi - 18\sqrt{3}$

b. $12\pi - 9\sqrt{3}$

c. $36\pi - 18\sqrt{3}$

d. $24\pi - 9\sqrt{3}$

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11 A sector OAB has radius 10 cm and angle $\theta = 0.8$ rad. A line perpendicular to OA is drawn from B to meet OA at C. Find the area of the shaded region bounded by arc AB and lines BC and AC.

a. $40 - 24.5 = 15.5\ \text{cm}^2$

b. $20.2\ \text{cm}^2$

c. $12.8\ \text{cm}^2$

d. $18.4\ \text{cm}^2$

12 In a circle of radius r, the area of a sector is numerically equal to the perimeter of the sector. What relation between r and θ must hold?

a. $\theta = \dfrac{4}{r-2}$

b. $\theta = \dfrac{2r}{r-2}$

c. $\theta = \dfrac{2}{r}$

d. $\theta = \dfrac{4}{r}$

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